

PROJECT PROPOSAL

Members:

Section: *Jasmine*

- *Winzy Rhoanne O. Costa*
- *Keane Rayleigh S. Riel*
- *Amara Niña C. Saludaes*

Project Title: *The Water Intake Questionnaire*

Problem Statement:

Many people forget to drink water sometimes, and often do not even know if they drank enough or exceeded the amount of water to drink for a day. One problem is how much they should drink actually, the classic “drink 8 glasses of water” everyday won’t work for everyone. Different people with different lifestyles would need different amount of water intake.

This is where our “The Water Intake Questionnaire” comes in. It calculates how much water the user has to drink depending on factors like weight, exercise frequency, sex, and age. This helps users know exactly how much water they need and if they reached their goal.

Planned Objectives:

The main goal of our project is to build a functional Python program that helps users keep track of their daily hydration habits. First, we aim to design a system that takes in a user's basic info, like their weight, age, sex, and how often they work out, and uses those specific metrics to calculate a personalized daily water goal. From there, we want to create a continuous loop where users can easily log how many cups they drink and what time they drank them throughout the day. At the end of the day, the program will calculate the final total and compare it to their goal, giving them a clear breakdown of whether they stayed hydrated or fell short.

Planned Features:

- *Enter user's name, their weight, exercise frequency, sex, water goal intake, and age.*
- *Show options for exercise frequency: A. Never B. Rarely C. Normally D. Frequently E. Always*
- *Calculate the base water goal in cups of the user by dividing their body weight in pounds by 16.*

- *Show options for biological sex: M. Male: Adds 3 extra cups to the goal for average muscle mass needs. F. Female: Uses the regular baseline amount.*
- *Shows rules based on age groups: Under 14: Sets small goal because they have smaller bodies. 14-59: Uses the standard baseline based on body weight. 60 and above: Sets a safe minimum goal to protect older adults from dehydration.*
- *Ask the user's water intake and the time of the drink inside a loop until they choose to end the day, to see if they reached their target or if they need to drink more.*
- *Use a variable to hold the total cups of water and compare it to the goal to see if the target was achieved at the end of the day.*

Planned Inputs and Outputs:

INPUTS:

- *Name*
- *Weight in pounds*
- *Age*
- *Sex*
- *Exercise Frequency*
- *How many cups did they drink within the hour*
- *Time of the day*
- *Answer to end the day (1 or 2)*

OUTPUTS:

- *Welcome message with the user's name*
- *The choices for exercise frequency and sex options*
- *The questions asking how much they drank and what time it is*
- *Number of cups needed for the day*
- *Total cups drank at the end of the day*
- *Final answer saying if they drank enough water today ("Yes" or "No")*

Logic Plan (Pseudocode):

START

PRINT "Hello, welcome to your very own water intake tracker! (Be grateful it's free)
Please state your name."

INPUT NAME

PRINT "Hi, " + NAME + "! Before we start, please enter your details."

```
PRINT "Your weight in pounds:"  
INPUT WEIGHT
```

```
PRINT "Your age:"  
INPUT AGE
```

```
IF AGE < 14 THEN  
    PRINT "This program isn't targeted towards your age range just yet!"  
END IF
```

```
PRINT "Your sex (Please enter Male or Female only):"  
INPUT SEX
```

```
PRINT "And how frequently you exercise (A. NEVER B. SELDOMLY C. SOMETIMES D.  
FREQUENTLY E. ALWAYS):"  
INPUT EXERCISE
```

```
RECOMMENDED_INTAKE = WEIGHT / 16
```

```
IF SEX = "MALE" AND AGE >= 14 AND AGE < 60 THEN  
    RECOMMENDED_INTAKE += 3  
ELSE IF SEX = "FEMALE" AND AGE >= 14 AND AGE < 60 THEN  
    RECOMMENDED_INTAKE = RECOMMENDED_INTAKE  
ELSE IF AGE >= 60 THEN  
    RECOMMENDED_INTAKE = 8  
END IF
```

```
IF EXERCISE = "A" THEN  
    RECOMMENDED_INTAKE = RECOMMENDED_INTAKE  
ELSE IF EXERCISE = "B" THEN  
    RECOMMENDED_INTAKE += 1  
ELSE IF EXERCISE = "C" THEN  
    RECOMMENDED_INTAKE += 2  
ELSE IF EXERCISE = "D" THEN  
    RECOMMENDED_INTAKE += 3  
ELSE IF EXERCISE = "E" THEN  
    RECOMMENDED_INTAKE += 4  
END IF
```

```
PRINT "Thank you for completely answering our questions! Come back when you drink  
water!"
```

```
NEW_QTY = 0  
CHOICE = 2 // Starts at 2 so the loop runs
```

```
WHILE CHOICE = 2 DO  
    PRINT "How many cups did you drink within the hour?"  
    INPUT QTY
```

```
PRINT "What time of the day is it?"  
INPUT HOUR
```

```
NEW_QTY += QTY
```

```
PRINT "Do you want to end the day? 1 for Yes, 2 for No."  
INPUT CHOICE  
END WHILE
```

```
PRINT "Number of cups needed: " + RECOMMENDED_INTAKE + ", Total cups drank: " +  
NEW_QTY + "."
```

```
IF NEW_QTY >= RECOMMENDED_INTAKE THEN  
    PRINT "Did you drink enough? Yes!"  
ELSE  
    PRINT "Did you drink enough? No! Try hydrating yourself!"  
END IF
```

```
END
```